

Practical exercises

— experimental and theoretical probability

A Cambridge insert: what the die should do, what it actually does, and why the two converge.

LESSON 164

1 × 40 MIN

MATEMATIKA 7, AMALIY MASHQLAR

S8 13

LESSON CLOCK

10'

12'

10'

6'

2'

Theoretical probability	10 min
The experiment	12 min
Why they differ	10 min
Expected numbers	6 min
Homework	2 min

LEARNING OBJECTIVES

- Calculate a theoretical probability by counting outcomes.
- Calculate an experimental probability from a frequency table.
- Explain why the two differ and what makes them close.
- Estimate the expected number of successes in a run of trials.

TEXTBOOK

National	pp. 495–498
Cambridge	Stage 8, pp. 126–134
Workbook	Exercise 13.4

01

Explanation

Theoretical probability

$$p = \frac{\text{favourable outcomes}}{\text{all outcomes}}$$

It is a counting problem — which is why combinatorics came first.

EVENT	FAVOURABLE	ALL	P
a six on one die	1	6	$\frac{1}{6} \approx 0.17$
an even number	3	6	$\frac{1}{2}$
a head on a coin	1	2	0.5
a total of 7 on two dice	6	36	$\frac{1}{6}$

EVERY PROBABILITY LIES BETWEEN 0 AND 1

An impossible event has $P = 0$, a certain one $P = 1$. An answer outside that range is a signal that the counting went wrong.

The experiment

Throw a die 60 times and tabulate. The **relative frequency** of a face is the number of times it appeared divided by the number of throws.

FACE	1	2	3	4	5	6
frequency	8	12	9	11	10	10
relative frequency	0.13	0.20	0.15	0.18	0.17	0.17

The theoretical value is $\frac{1}{6} \approx 0.17$ for every face; the experiment scatters around it.

THE FREQUENCIES MUST ADD TO THE NUMBER OF THROWS

$8 + 12 + 9 + 11 + 10 + 10 = 60 \checkmark$, and the relative frequencies add to 1. Both checks take a moment and catch a miscount.

Why they differ

THROWS	SIXES	RELATIVE FREQUENCY	DISTANCE FROM 0.17
10	3	0.30	far
60	10	0.17	close
600	104	0.173	closer

A LONG RUN OF SIXES DOES NOT MAKE THE NEXT SIX LESS LIKELY

The die has no memory. Relative frequency settles down because later throws swamp the early ones, not because the die corrects itself.

Expected numbers

expected number = $P \times$ number of trials

EXPERIMENT	WORKING	EXPECTED
sixes in 60 throws	$\frac{1}{6} \cdot 60$	10
heads in 50 tosses	$0.5 \cdot 50$	25
evens in 90 throws	$0.5 \cdot 90$	45
totals of 7 in 180 throws of two dice	$\frac{1}{6} \cdot 180$	30

“EXPECTED” MEANS ON AVERAGE, NOT EXACTLY

Ten sixes in sixty throws is the average outcome, not a promise. Getting eight or thirteen is ordinary; getting none at all would be evidence that the die is biased.

02 Worked examples

EXAMPLE 1 A die is thrown 60 times and a six appears 13 times. Compare the experimental and theoretical probabilities.

① Experimental: $\frac{13}{60} \approx 0.22$.

② Theoretical: $\frac{1}{6} \approx 0.17$.

③ The experimental value is a little higher.
Ordinary variation in 60 throws.

ANSWER

0.22 against 0.17

EXAMPLE 2 How many heads would you expect in 50 tosses of a fair coin?

① $P = 0.5$

② $0.5 \cdot 50$

③ $= 25$
On average, not exactly.

ANSWER

25

EXAMPLE 3 Two dice are thrown. Find the probability of a total of 7.

① All outcomes: $6 \cdot 6 = 36$.

② Favourable: $(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)$ — six of them.

③
$$P = \frac{6}{36} = \frac{1}{6}$$

ANSWER

$$\frac{1}{6}$$

03 **Interactive model**

Pool the whole class's throws on the board: each pupil's twenty look erratic, and the combined six hundred sit visibly close to a sixth.

Theoretical, experimental, expected

P (a six on one die) is:

$$\frac{1}{2}$$

$$\frac{1}{3}$$

$$\frac{1}{6}$$

$$\frac{1}{12}$$

P (an even number) is:

$$\frac{1}{6}$$

$$\frac{1}{3}$$

$$\frac{1}{2}$$

$$\frac{2}{3}$$

A six in 13 of 60 throws gives an experimental probability of:

0.13

0.17

0.22

0.60

Expected sixes in 60 throws:

6

10

13

30

Expected heads in 50 tosses:

10

25

50

5

Relative frequencies of all faces add to:

0

0.5

1

6

After five sixes in a row, the next throw gives a six with probability:

$$\frac{1}{6}$$

less than $\frac{1}{6}$

more than $\frac{1}{6}$

0

Experimental values approach theoretical ones when:

the die is heavier

more trials are made

fewer trials are made

never

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Probability	Ehtimollik	Вероятность
Theoretical	Nazariy	Теоретическая
Experimental	Tajribaviy	Экспериментальная
Relative frequency	Nisbiy chastota	Относительная частота
Trial	Sinov	Испытание
Fair	Bir jinsli	Правильный
Biased	Nosoz	Смещённый
Expected number	Kutilayotgan soni	Ожидаемое число

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Theoretical probability is found by:

experiment

counting outcomes

guessing

measuring

Experimental probability is found by:

counting outcomes

dividing frequency by trials

multiplying

estimating

Every probability lies between:

0 and 1

-1 and 1

0 and 100

any values

Expected sixes in 180 throws:

$P(\text{total } 7 \text{ on two dice})$ is:

More trials make the experimental value:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $P(\text{a six on one die})$

ANSWER $\frac{1}{6}$

2. $P(\text{an even number})$

ANSWER $\frac{1}{2}$

3. $P(\text{a head})$

ANSWER 0.5

4. $P(\text{a number over 4})$

ANSWER $\frac{1}{3}$

5. Expected heads in 50 tosses

ANSWER 25

6. Expected sixes in 60 throws

ANSWER 10

7. Relative frequency of 12 in 60

ANSWER 0.20

Medium · the standard question

7 PROBLEMS

1. A six in 13 of 60 throws: the experimental probability

ANSWER ≈ 0.22

2. $P(\text{total } 7 \text{ on two dice})$

ANSWER $\frac{1}{6}$

3. $P(\text{total } 12 \text{ on two dice})$

ANSWER $\frac{1}{36}$

4. Expected totals of 7 in 180 throws

ANSWER 30

5. Do relative frequencies add to 1?

ANSWER Yes

6. After five sixes, $P(\text{six})$ is

ANSWER $\frac{1}{6}$

7. Expected evens in 90 throws

ANSWER 45

Hard · stretch and reason

7 PROBLEMS

1. $P(\text{at least one six on two dice})$

ANSWER $\frac{11}{36}$

2. $P(\text{exactly one six on two dice})$

ANSWER $\frac{10}{36} = \frac{5}{18}$

3. $P(\text{at least one head on three coins})$

ANSWER $\frac{7}{8}$

4. A die gives 40 sixes in 60 throws: what would you conclude?

ANSWER It is very likely biased

5. Expected sixes in 600 throws

ANSWER 100

6. Why does relative frequency settle down?

ANSWER Later trials outweigh the early variation

7. $P(\text{a total under 5 on two dice})$

ANSWER $\frac{6}{36} = \frac{1}{6}$

07 Homework

Homework — 5 tasks

Record every throw as you make it; filling the table afterwards from memory is not an experiment.

1. Throw a die 60 times, tabulate the results and find each relative frequency.
2. Compare your six values with $\frac{1}{6}$ and comment.
3. Find $P(\text{a total of } 7)$ and $P(\text{a total of } 2)$ for two dice.

4. How many heads would you expect in 80 tosses of a coin?
5. Explain why a run of five heads does not change the probability of the next toss.